

Task 10 - Firewall Configuration & Testing

Chapter 1: Introduction to Firewalls

A firewall is a network security mechanism that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a protective barrier between a trusted internal system and untrusted external networks.

Firewalls help prevent unauthorized access, reduce exposure to attacks, and enforce security policies by filtering network traffic at the system or network level.

Chapter 2: Need for Firewall Protection

Modern systems often run multiple network services that listen on different ports. Without firewall protection, these services remain exposed to attackers, increasing the system's attack surface.

Firewall protection is necessary to:

- Restrict unauthorized access
- Protect sensitive services
- Reduce risk from network-based attacks
- Enforce access control policies

An initial network scan without firewall rules highlights the importance of implementing firewall protection.

Chapter 3: Firewall Installation and Setup

In this task, **UFW (Uncomplicated Firewall)** was used to configure firewall rules on Kali Linux. UFW provides a simplified interface for managing firewall rules while internally using iptables.

UFW operates as a **stateful firewall**, meaning it tracks the state of active connections and allows return traffic for established sessions, making it more secure than stateless filtering.

```

Installing:
  ufw

Suggested packages:
  rsyslog

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 474
  Download size: 169 kB
  Space needed: 880 kB / 2007 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 ufw all 0.36.2-9 [169 kB]
Fetched 169 kB in 1s (157 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ufw.
(Reading database ... 400918 files and directories currently installed.)
Preparing to unpack .../archives/ufw_0.36.2-9_all.deb ...
Unpacking ufw (0.36.2-9) ...
Setting up ufw (0.36.2-9) ...
Creating config file /etc/ufw/before.rules with new version
Creating config file /etc/ufw/before6.rules with new version
Creating config file /etc/ufw/after.rules with new version
Creating config file /etc/ufw/after6.rules with new version
update-rc.d: We have no instructions for the ufw init script.
update-rc.d: It looks like a non-network service, we enable it.
Created symlink '/etc/systemd/system/multi-user.target.wants/ufw.service' -> '/usr/lib/systemd/system/ufw.service'.
Processing triggers for kali-menu (2025.2.0) ...
Processing triggers for man-db (2.13.0-1) ...

(ravnyx@hope)-[~]
$ ufw --version
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.

(ravnyx@hope)-[~]
$

```

Chapter 4: Enabling the Firewall

Before enforcing firewall rules, the firewall must be enabled. Enabling the firewall activates rule enforcement and begins monitoring network traffic.

Once enabled, the firewall applies default policies and starts filtering traffic based on configured rules, ensuring that only authorized connections are permitted.

```

(ravnyx@hope)-[~]
$ sudo ufw status
Status: inactive

(ravnyx@hope)-[~]
$

```

```

(ravnyx@hope)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

```

Chapter 5: Firewall Rule Configuration

Firewall rules were configured to control incoming and outgoing traffic. The following rules were applied:

- Allow SSH (Port 22) for secure remote access
- Allow HTTP (Port 80) for web services
- Allow MySQL (Port 3306) for database access
- Deny all other incoming traffic by default

This configuration follows the **principle of least privilege**, allowing only essential services.

```
(ravnyx@hope)-[~]
$ sudo ufw allow ssh
Rule added
Rule added (v6)

(ravnyx@hope)-[~]
$ sudo ufw allow http
Rule added
Rule added (v6)

(ravnyx@hope)-[~]
$ sudo ufw allow mysql
Rule added
Rule added (v6)

(ravnyx@hope)-[~]
$
```

Chapter 6: Firewall Rule Verification

After configuring firewall rules, verification is required to ensure correct enforcement. Firewall rule verification confirms:

- Active firewall status
- Default deny policy for incoming traffic
- Explicitly allowed services

Verification helps prevent misconfigurations that could lead to unintended access.

```
(ravnyx@hope)-[~]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
3306/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
3306/tcp (v6) ALLOW IN Anywhere (v6)

(ravnyx@hope)-[~]
$
```

Chapter 7: Testing Firewall Connectivity

Firewall rules were tested using **Nmap** network scanning. Scans performed after firewall configuration showed that only permitted ports remained accessible, while other ports were filtered or blocked.

This testing confirms the effectiveness of the firewall in controlling network access.

```
(ravnyx@hope)-[~]
$ nmap 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-30 13:05 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

(ravnyx@hope)-[~]
$
```

Chapter 8: Logging and Monitoring

Firewall logging was enabled to monitor network activity. Logs record:

- Allowed connections
- Blocked access attempts

Logging is essential for detecting suspicious behavior, analyzing incidents, and maintaining security visibility.

```
(ravnyx@hope)-[~]
$ sudo ufw logging on
Logging enabled
```

Chapter 9: Blocking Malicious Traffic

Firewalls can block traffic from specific IP addresses to prevent unauthorized or suspicious access. In this task, a sample IP address was blocked to demonstrate firewall-based access restriction.

IP blocking is useful for responding to detected threats or preventing repeated unauthorized attempts.

```
(ravnyx@hope)-[~]
$ sudo ufw deny from 192.168.1.100
Rule added
```

Chapter 10: Impact and Limitations of Firewall

Firewall implementation significantly improves system security by:

- Reducing exposed services
- Enforcing access control
- Preventing unauthorized network access

However, firewalls have limitations. They cannot prevent:

- Application-level vulnerabilities
- Insider threats
- Social engineering attacks

Firewalls must be combined with other security measures for comprehensive protection.

Chapter 11: Conclusion

This task demonstrated the practical implementation of firewall configuration and testing using UFW. By defining rules, verifying configurations, testing connectivity, and enabling logging, the task highlighted the role of firewalls in enhancing system security and controlling network traffic.